UNIVERSITY of RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY
SCHOOL OF ICT

DEPARTMENTS: COMPUTER AND SOFTWARE ENGINEERING
CAMPUS: NYARUGENGE AND GAKO

END OF SEMESTER EXAMINATION -ACADEMIC YEAR 2025-2026

YEAR: 2 SEMESTERS: I

MODULE CODE and NAME: COE2161 Advanced computer programming in C++

DATE: 14/01/2026

TIME: 2hours

MAXIMUM MARKS = 50

INSTRUCTIONS

1. This paper contains **FOUR (4)** questions.
2. **Answer THREE (3) Questions only:**
   Question ONE (1) from Section "A" is Compulsory and Answer any TWO (2) Questions from Section "B"
3. No written materials allowed.
4. Do not forget to write your Registration Number.
5. Write all your answers in the booklet provided
6. You may use non-programmable calculators
7. Do not write any answers on this questions paper.
8. **Start each question in a NEW page**

CamScanner

SECTION A: COMPULSORY

Question 1: (20 Marks)

a. Explain the Advantages and disadvantages of using C++ Standard Template Library. **(6 marks)**

b. With help of syntax, Define Pointers and their functions in C++. **(4 Marks)**

c. John Doe is developing a Library Management System using C++. The system manages different types of library items such as books, magazines, and digital resources. All items share common properties like id and title, but each item behaves differently when accessed or borrowed. John designs the system using object-oriented principles to allow different item types to be handled uniformly while maintaining their specific behaviors. He also uses templates to create reusable components that work with various data types. To manage collections of items efficiently, John relies on the C++ Standard Template Library, including containers and algorithms. During development, John must make design choices that affect type safety, runtime behavior, and performance.

Choose the most appropriate answer based on the shared information

i. John stores different types of library items in a single container and calls the same function on each object, expecting the correct behavior based on the actual object type. **(2 Marks)**

A. Defining the function in each class without inheritance
B. Declaring the function as virtual in a common base class
C. Overloading the function name in derived classes
D. Using a template function instead of inheritance

ii. John wants to ensure that once an item's id is assigned, it cannot be modified directly from outside the class but can still be read when needed. **(2 Marks)**

A. Declare id as a public member
B. Declare id as protected and access it in derived classes
C. Declare id as private and provide a public getter
D. Declare id as a global constant

iii. John creates a class template that can store different types of library items without rewriting the class for each type. **(2 Marks)**

A. It allows runtime selection of object types
B. It enables code reuse for different data types
C. It automatically provides polymorphic behavior
D. It improves memory management by default

CamScanner

iv. John creates a base class LibraryItem and derives Book and Magazine from it. He wants Book and Magazine to reuse the common data and behavior defined in LibraryItem. **(2 Marks)**

A. Encapsulation
B. Inheritance
C. Abstraction
D. Overloading

v. John has a collection of items stored in a std::vector and wants to arrange them in ascending order based on their titles **(2 Marks)**.

A. Use std::sort with a custom comparison function
B. Use std::find_if repeatedly
C. Use std::binary_search
D. Use std::count_if

SECTION B: CHOOSE ONLY 2 QUESTIONS

Question:2 (15 Marks)

a. **Write the output produced by these program (5 Marks)**

a. Write the output produced by these program (5 Marks)

```
#include <iostream>
using namespace std;
int main() {
    int sum = 0;
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= i; j++) {
            if (j % 2 == 0) {
                sum += j;
            } else {
                sum -= j;
            }
        }
        cout << sum << " ";
    }
    return 0;
}
```

b. Write the output produced by these program (5 Marks)

```
#include <iostream>
using namespace std;
void printLine(int i, int n) {
    if (i > n) return;
    cout << i;
    printLine(i + 1, n);
}

void printPattern(int n) {
    if (n == 0) return;
    printPattern(n - 1);
    printLine(1, n);
    cout << endl;
}

int main() {
    printPattern(5);
    return 0;
}
```

CamScanner

c. **Write the output produced by these program (5 Marks)**

```
1  using namespace std;
2  int main() {
3      int n = 5;
4      for (int i = 1; i <= n; i++) {
5          // Print spaces
6          for (int j = 1; j <= n - i; j++) {
7              cout << " ";
8          }
9          for (int j = 1; j <= i; j++) {
10             cout << j;
11         }
12         cout << endl;
13     }
14
15     return 0;
16 }
```

Question 3: (15 Marks)

Write a single C++ program that demonstrates the following:
a. A function 'incrementByValue' that increments an integer by 1 using call by value.
b. A function 'incrementByReference' that increments an integer by 1 using call by reference.
c. A function 'swapNumbers' that swaps two integers using call by reference.

In main():
- Declare an integer 'num = 5', call both increment functions, and print 'num' after each call.
- Declare two integers 'a = 10' and 'b = 20', call 'swapNumbers(a, b)', and print the swapped values.
- Add brief comments explaining the differences in behavior.

Expected Output:
After call by value: 5
After call by reference: 6
After swapping: a = 20, b = 10

Question 4: (15 Marks)

John Doe is developing a **small Animal application** in C++.
The application should allow different types of animals (like dogs and cats) to make sounds.
John also wants the program to decide which animal sound to make based on user input.

CamScanner

wrote the following program, but it contains several errors.

```
#include <iostream>
using namespace std;

class Animal {
public:
    void makeSound() {
        cout << "Some generic animal sound" << endl;
    }
};

class Dog : Animal {
public:
    void makesound() {
        cout << "Woof Woof" << endl;
    }
};

class Cat : Animal {
public:
    void makeSound() {
        cout << "Meow Meow" << endl;
    }
};

int main() {
    string type;
    cout << "Enter animal type: ";
    cin >> type;

    Animal *a;

    if (type = "dog") {
        a = new Dog();
    } else if (type == "cat") {
        a = new Cat();
    } else {
        a = new Animal();
    }

    a->makeSound();

    return 0;
}
```

TASK

Identify all errors in the code, explain what is wrong, suggest how it can be fixed and you do not need to rewrite the entire program.

On line 100: Error – Declaring int x = 4.6 is not allowed because an int cannot store decimal values.

Fix: Assign an integer value instead, e.g., int x = 4;

END

CamScanner